UMPIREAL

Traffic Congestion The Energy Blackhole Of The Modern World

Stop Start Traffic Wastes Up To 5,000 TWH Of Energy Per Year Globally In Comparison The Entire World's Nuclear Power Station Fleet Produces Roughly 2,600 TWH Per Year.

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UMPIREAL

Business Model 1: Umpireal House of Data

Traffic Flow Preservation with Integrated Real Estate: Sovereign AI Edge Infrastructure · IBM + NVIDIA + BYD · 8 GWh Energy Offset Per Node

INVESTOR PRESENTATION · APRIL 2026

Stop-start driving forces cars to burn 67% more fuel at every junction—while for HGVs, that waste spikes to 87%. This massive energy leak heats our cities, drains logistics fleets, and overloads the grid. For decades, this has been an overlooked “stopping tax" on urban motion.

NODE CAPEX	YR 1 EBITDA	PAYBACK	20-YR EBITDA
€5.79M	**€2.58M**	**~ 32 months**	**~ €240M**
ex racks + grid	87.2% EBITDA · €2.24M net to SPV · standard 4-rack node	4 racks standard · scales to flagship	~24×Per Node

Every Junction Is a Net Energy Burden

PENALTY 1 — BRAKE HEAT

When a vehicle stops at a junction, kinetic energy built up by the engine is converted entirely to brake heat and friction. It enters the urban atmosphere as waste heat. It cannot be recovered. It is simply gone.

PENALTY 2 — RE-ACCELERATION

The engine then re-accelerates from standstill at its lowest efficiency point — burning maximum fuel to restore the momentum that was just discarded. This is the most expensive moment in any journey.

Cars burn

67% MORE FUEL

at every stop vs cruising

HGVs burn

87% MORE FUEL

at every stop vs cruising

Wasted energy

8 GWh / yr

per busy junction

CO_2 emitted

2,000 t / yr

per junction — needlessly
Or 1.1 Million m3 of CO_2

For 30 years, Euro 1 to Euro 6 policy improved engines. Nobody applied a single standard to the road itself. The junction stayed broken. UMPIREAL is a leap to euro 10 in one bound.

THE BÉAR CROSSING

The BÉAR Crossing is a road junction geometry that routes pedestrians safely underneath the road so vehicles never stop. No stopping = no kinetic energy lost to brake heat. No re-acceleration from standstill. No urban heat pulse.

TRAFFIC

Vehicles never stop

Traffic preserves momentum through the junction. Stop-start friction is permanently eliminated. Every vehicle, every crossing, every day.

PEDESTRIANS

Cross safely underneath

A dedicated pedestrian underpass runs beneath the raised road. Full DDA compliance, lighting, wayfinding. Safer than a surface crossing at any traffic speed.

REAL ESTATE

Space created below

The geometry creates enclosed, structurally sound, weatherproof space beneath the crest. Modular, scalable, deployable at any junction. The road pays for the building.

This is not carbon capture. There is nothing to capture — because thanks to Umpireal, it was never burned.

What Lives Beneath the Road

Standard node: 9 rooms · 620 m² technical + 180 m² hospitality · single 500 kW commercial grid connection · shell built to flagship structural spec throughout.

Room 1

DATA CENTRE

4 × NVIDIA VR NVL72 (Vera Rubin)
288 GPUs · 14.4 EFLOPS FP4
460 kW IT load · 7 racks total
Direct liquid cooled — 190 kW/rack
Fabric + DPU + mgmt racks included
Scales to 8, 16, 32 racks — no rebuild

Room 2

BESS STORAGE

BYD Blade LFP · 7MWh · 1 room
7 MWh · 3.5 MW · LFP chemistry
Grid services + arbitrage
€280–420K/yr grid services

Room 3

M&E PLANT

CDUs · pump sets
Plate heat exchangers
LV switchgear · UPS
500 kW grid connection

Room 4

BACKUP / FLEX

ATS · generator stub
Future BESS string or commercial let
rough-in pre-installed

Room 5

COMMERCIAL / CIVIC

Café · fast food restaurant
EV fast charging
Co-working / NOC
Retail · office · civic space
High-footfall location

STRUCTURAL SHELL 9-room node · 620 m² technical · 180 m² hospitality · Design life: 100+ years · Slab-on-grade: 1,875 kg/m² VR NVL72 rated · Upgrades to 8/16/32 racks require no structural changes

CAPITAL COST

€5.79M Colocation Tenant Ready

BREAKDOWN

Structural shell (precast + install + civils)	€1.89M	33%
Above-ground road works + civils + TM	€0.8M	9%
Data room fit-out — DLC-ready, IaaS landlord scope (4× NVL72)	€1.04M	18%
BESS — 7MWh BYD Blade LFP · 1 room (€120/kWh installed)	€0.975M	17%
Cooling + M&E plant + backup power	€0.530M	9%
Hospitality + commercial tenancy fit-out	€0.472M	8%
Professional fees + contingency (18%)	€0.884M	15%
TOTAL NODE CAPEX (ex NVL72 racks + grid)	€5.79M	

VS CONVENTIONAL DC

Conventional build

€41–46M

Source: Savills Ireland / Turner & Townsend 2023 · €8.4–10M/MW
Includes: land, shell, electrical, HVAC, cooling, fire, fitout

Umpireal node

€5.79M

Includes road, ramps, utilities, BESS, district heating, security and traffic monitoring

86% CHEAPER — MORE CAPACITY

Four Uncorrelated Revenue Streams

1 — COLOCATION
92% EBITDA

€2.0–2.5M/yr

$500/kW/mo · 4 × VR NVL72 · 460 kW · 80% occ.

Tenant brings own GB200 racks. Umpireal sells conditioned space, power and liquid cooling. Energy is a pass-through. No rack capex, no depreciation, no utilisation risk.

2 — BESS GRID SERVICES
High margin

€280–420K/yr

Conservative: €280K · Upside: €420K

Arbitrage (€2.5–3.5M) + Frequency Response FCR/aFRR (€3.0–4.5M) + Capacity Market (€0.5–1.0M). The 3.5 MW / 7 MWh BYD Blade LFP node qualifies for high-value Primary Frequency Response contracts. LFP chemistry ideal for enclosed underground deployment.

3 — DISTRICT HEATING
Pure upside

€102K/yr

~260 homes · 2.6 GWh thermal · GPU waste heat

GPU liquid cooling waste heat recovered and exported to local district heating network. At 460 kW IT load, ~260 homes served. Scales with each rack upgrade.

4 — CARBON CREDITS
Attributable

€104K/yr

2,088 t CO_2 offset · EU ETS @ €50/t

The fuel not burned at the junction (2,000 t) plus grid carbon displaced by BESS (88 t) = 2,088 t of attributable CO_2 offset per node per year at standard configuration.

Plus: IP licence fee €160K (yr 1) · Commercial tenancy €150K/yr · EV charging €30K/yr · Node upgrades to 8/16 racks add proportional colocation revenue on same infrastructure

Year 1 P&L — Colocation Model

REVENUE		
Colocation IaaS (460 kW · $500/kW/mo · 80% occupancy · 4 × VR NVL72)		€2.01M
BESS grid services + arbitrage (7MWh BYD Blade · conservative)		€350K
District heating (~260 homes, 2.6 GWh thermal)		€102K
EU ETS carbon credits (2,088 t @ €50/t)		€104K
Hospitality + tenancy + EV charging + IP licence (yr 1)		€390K
TOTAL GROSS REVENUE		€2.956M
COSTS		
Energy + IT rack capex (pass-through / tenant-owned — zero net cost)		€0
Node opex (maintenance, insurance, monitoring staff)		€380K
EBITDA (pre-distribution)	€2.576M	87.1%
Ground rent to council (€250/m² × 770m²)		−€193K
Commercial rates (council levy)		−€150K
Net to Umpireal SPV		€2.233M

20-YEAR MODEL

Three Upgrade Cycles. Compounding Returns.

YEAR 3 — RUBIN ULTRA NVL576
+25% colocation rate · tenant upgrades to Rubin Ultra · 600kW/rack · operator uplifts rent

YEAR 7 — FEYNMAN GENERATION
+20% colocation rate · Feynman generation · same shell · new compute standard

YEAR 12 — GEN 5 + BESS REFRESH
+20% rate · BESS modules €1.9M · enclosures/switchgear kept

Yr	Colo Rev	Ancillary	Total Rev	Opex	BESS Rep.	EBITDA	Margin	Cumulative
1	€2.01M	€0.56M	€2.57M	€0.55M	—	€2.58M	85%	€2M
2	€2.07M	€0.58M	€2.65M	€0.39M	—	€2.26M	85%	€5M
3	€4.14M	€0.60M	€4.74M	€0.42M	—	€4.32M	95%	€11M
5	€8.28M	€0.64M	€8.92M	€0.48M	—	€8.44M	95%	€29M
7	€9.94M	€0.68M	€10.62M	€0.51M	—	€10.11M	95%	€48M
10	€19.9M	€0.74M	€20.6M	€0.55M	—	€20.1M	95%	€93M
12	€20.5M	€0.79M	€21.3M	€0.58M	€1.9M	€18.8M	88%	€134M
15	€24.6M	€0.86M	€25.5M	€0.64M	—	€24.8M	97%	€203M
20	€26.0M	€1.00M	€27.0M	€0.74M	—	€28.8M	97%	~€240M

Yr 1 EBITDA
€2.58M

Yr 20 EBITDA
€28.8M

Cumulative 20yr
~€240M

MOIC on node
~24×

DEPRECIATION

The Shell Is Permanent. The Technology Is a Subscription.

Asset	Cost	Life	Annual Depr.	Risk	Cycle
Precast shell + structural civils	€2.1M	120 yrs	€9,600/yr	Negligible	Never — design life outlasts all technology
Above-ground road works	€0.8M	30 yrs	€47,000/yr	Very low	Yr 30 — standard road maintenance cycle
Data room fitout	€1.0M	20 yrs	€50,000/yr	Low	Yr 20 — outlasts 5+ rack generations
BESS — 7MWh BYD Blade LFP (2 rooms)	€1.9M	12 yrs	€158,000/yr	Medium	Yr 12 — modules €1.9M, keep enclosures & switchgear
Cooling + district heating	€0.7M	20 yrs	€35,000/yr	Low	Yr 20 — alongside data room fitout
Monitoring + security	€0.9M	10 yrs	€90,000/yr	Low	Yr 10 — technology refresh €0.30M
IT racks VR NVL72 (Vera Rubin)	€0 to operator	3–4 yrs	€0 to operator	Zero	Tenant owns, depreciates and replaces
BLENDED OPERATOR DEPRECIATION (years 1–5)					**~€503K/yr**

5.4% of capex/yr — extraordinarily low for an asset generating 95%+ EBITDA margins. After year 5, blended depreciation drops to ~€413K/yr as soft costs fully amortise.

VR NVL72 STRUCTURAL ADVANTAGE Each VR NVL72 rack uses the same Oberon rack architecture as GB300 — same footprint, same 1,875 kg/m² floor load, now with direct liquid cooling. The Umpireal precast slab-on-grade shell supports this load natively. 80% reduction in rack airflow vs GB300 means lower ventilation infrastructure cost. The building improves with every compute generation — the shell never needs modification.

BS 8004 Compensated Foundation Analysis

Component	Basis	kN/m²
	SECTION 1 — GROSS LOADS AT BASE (77 M² FLOOR AREA)	
Precast shell 75 m³ @ 24.5 kN/m³ ÷ 77 m²	Dead	23.9
32 × VR NVL72 racks 49,700 kg ÷ 77 m²	Dead	6.3
Road formation 600 mm avg 21 kN/m³	Dead	12.6
Hardcore 300 mm crushed stone 20 kN/m³	Dead	6.0
100 t live traffic 2 × 44t HGVs, direct on roof slab	Live	13.0
Gross total at base		**61.8**
	SECTION 2 — EXCAVATION CREDIT ($Q_{nu} = Q_u - \Gamma DF$)	
Soil displacement credit 2.3 m × 1.9 t/m³ × 9.81 = 42.8 kN/m²	BS 8004 γDf	−43.0
Net bearing pressure at foundation level		**18.8**
	SECTION 3 — LOAD DISPERSION TO SUBGRADE	
200 mm XPS rigid board — negligible spread	—	—
50 mm fines φ = 25°, spread = 23 mm each side	φ = 25°	—
300 mm hardcore 3" down φ = 45°, spread = 300 mm each side	φ = 45°	—
Dispersed area at subgrade (10.0 + 0.60) × (7.7 + 0.60) = 87.98 m²	Area	87.98 m²
Net bearing pressure at subgrade — 18.8 × 77 ÷ 87.98		**16.5 kN/m²**
	SECTION 4 — BS 8004 BEARING CAPACITY CHECK (NET ALLOWABLE VS 16.5 KN/M²)	

BS 8004 Bearing Capacity — Ground Type Check

Ground type	BS 8004 net allowable	Safety factor	Status
Dense gravel / sand-gravel	600 kN/m²	36.4×	Pass
Stiff boulder clay	300 kN/m²	18.2×	Pass
Stiff clay	150 kN/m²	9.1×	Pass
Firm clay	75 kN/m²	4.5×	Pass
Soft clay	40 kN/m²	2.4×	Ground improvement
Very soft clay / made ground	<25 kN/m²	<1.5×	Not suitable

The BEAR Crossing colocation model is structurally and geotechnically viable. The compensated foundation principle confirms that the node imposes similar net loads to the road it sits beneath. Any challenges that arise beyond this will be site-specific and resolved through standard civil engineering practice.

PUBLIC BENEFIT

8 GWh Preserved. 2,000 Tonnes Avoided. Per Junction. Per Year.

8 GWh

Fuel energy preserved

Stays in tanks — never burned

2,000 t

CO_2 avoided

Equivalent to planting 200,000 trees/yr

€1.80M

Fleet fuel saving

~900,000 litres @ €2.00/L

60,833 hrs

Commuter time saved

€1.09M/yr value @ €18/hr

1.34 GWh

Urban heat not generated

Junction heat pulse eliminated

~280 homes

District heating served

GPU waste heat recovered

Umpireal is the ***Euro 10 for road infrastructure*** — skipping straight from zero performance standard to best-in-class in a single move. Every junction is a policy gap. Every BÉAR Crossing closes it.

IBM + NVIDIA + BYD — The Three-Layer Stack

At 1,000 nodes: IBM + NVIDIA deploy sovereign AI at urban scale · BYD ships **€1.05T** of VR NVL72 compute. BYD ships **€1.9B** of BESS units. Addressable market: **~5 million viable junctions globally.**

NVIDIA

VR NVL72 (Vera Rubin) · 32 racks per node · 6,080 kW · IBM watsonx + Red Hat AI Factory

Why NVIDIA + IBM:

· The GB200 needs liquid cooling — 95% of buildings can't support it
· Umpireal is 80% cheaper per MW than conventional DC build
· IBM Sovereign Core + VR NVL72 = regulated AI that never leaves the jurisdiction
· Slab-on-grade shell natively supports 1,875 kg/m² rack load

What we want:

IBM Ventures co-investment · VR NVL72 preferred partner
Joint pilot node · Metropolis / sovereign AI introductions

BYD ENERGY STORAGE

Blade LFP · 7MWh per node · 2 rooms · €1.9M per node · safest chemistry for enclosed nodes

Why BYD:

· Blade LFP — safest chemistry for enclosed underground nodes (nail penetration certified)
· Urban road infrastructure = new channel BYD doesn't have
· 7MWh per node · 3.5MW · 6,000+ cycle life — ideal for high-frequency grid services
· BYD operates in 70+ countries — accelerates global rollout

What we want:

Preferred partner + volume pricing · Joint pilot node
Volume pricing · Joint pilot node · Access to BYD's 70-country government/utility network

The Under-Crest Space — Configured to Location

The BÉAR Crossing creates enclosed, structurally sound real estate at every junction. The configuration is entirely location-driven — from a single café to a full commercial frontage with housing above.

Fast Food & QSR

McDonald's · Burger King · KFC · Subway

Captive high-footfall audience — commuters who no longer stop at the light but choose to pull in. A junction that never stops traffic is a franchise operator's dream location. Landlord rent: €120–180K/yr per unit.

Key Worker Housing

Nurses · teachers · emergency services

Modular residential units above or alongside the crossing — built in the infrastructure that's already there. No greenfield land. No planning battle. Lower delivery cost, faster build time, and a commute that's permanently faster.

Student Accommodation

Urban campuses · city centre locations

High-density, affordable student housing units integrated into junction infrastructure near universities. Solves the student housing crisis at the junction level — where students actually need to be.

Café & Hospitality

Coffee · bakery · convenience retail

The basic 5-room node includes a café unit. Scales to full hospitality frontage on larger nodes. Dwell time at the junction is now a commercial opportunity — not a traffic management problem.

Office & Civic Space

Co-working · community · services

Local authority services, health clinics, community hubs and co-working space embedded in transport infrastructure. Reduces car journeys by bringing services to the junction rather than the other way around.

EV Charging Hub

Tier 2–3 nodes · forecourt model

EV charging integrated with the BESS room — buy cheap overnight grid power, sell fast charging during the day. The junction that never stops traffic becomes the charging stop people choose to make.

The node scales with demand. Basic 5 rooms = 350 m². Each additional room adds ~€230K capex and opens another commercial use. Longer nodes (10, 15, 20+ rooms) accommodate full commercial frontage including drive-through restaurants — as shown in the render.

The Total Carbon Dividend

Heading: Beyond Net-Zero: The Carbon-Negative Infrastructure Anchor

Total Annual Carbon Impact per Node:

Offset Category	Metric	Trees Equivalent
Traffic Momentum	1,952 Tonnes CO2	88,746 Trees
District Heating	1,500 Tonnes CO2	68,181 Trees
BESS Grid Balancing	250 Tonnes CO2	11,363 Trees
TOTAL ANNUAL IMPACT	**3,702 Tonnes CO2**	**168,290 Trees**
**Estimated based on displacement of peak fossil-fuel "Peaker" plants.*		

€20M Seed Round — First Colocation Node. Then Scale.

Alan Ledwith

Founder & Inventor, Umpireal

Civil engineering · Construction management · Renewable energy · Serial founder
Bootstrapped €950K since April 2022. All IP, financial models, product design, market development. Solo.

WHAT IS DONE

- ✓ Geometry invented, IP filed (Application 2025/0540)
- ✓ Full node specification — all three embodiments
- ✓ Complete capex model from real supplier quotes
- ✓ Revenue model verified against Savills benchmarks
- ✓ 20-year financial model with upgrade cycles
- ✓ Strategic partner analysis — NVIDIA, BYD, Microsoft
- ✓ All components TRL 9 — innovation is in configuration

USE OF FUNDS — €20M SEED · 17–20% EQUITY

First node — pilot + site contingency
€7.29M · €5.79M build + €1.5M learning, delays & site contingency

HQ office + engineering lab + R&D
€1.50M · fit-out, equipment, node variant & turbogenerator R&D

Team — engineering, commercial, finance, operations
€3.00M · 15–20 technical and commercial hires

Global rollout — 3 jurisdictions, legal & IP
€2.75M · UK/EU/US pipeline · PCT filing · partner engagement

Sales, BD, exhibitions, advertising & brand
€1.50M · trade shows, events, digital media, branding

Founder salary + opex + working capital + contingency
€3.96M · €230K/yr founder · 24-month runway · working capital

This is not a proof of concept raise. The concept is proven. The components exist. The first node is a commercial asset. The raise builds the team and deploys the first node — then replicates.

UMPIREAL

Transport Infrastructure That Pays For Itself

Stop-start driving forces cars and HGVs to burn **67% and 87% more fuel** at every road junction compared to cruising. Decarbonising transport isn't just about the engine anymore. ***It's now about the road.***

Umpireal is the world's first Net-Positive Infrastructure Anchor — preserving **8 GWh** of energy and avoiding **2,000 tonnes of CO_2** per junction per year — then generating **€2.58M EBITDA in year one** from the space it creates beneath the road.

SEED RAISE
€20M

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"Umpireal — One Golden Goose. Infinite Golden Eggs. At Every Busy Crossing."

UMPIREAL

Urban Traffic Momentum Preservation
With Combined
Infrastructure Derived Real Estate

THANK YOU

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